

Phase transition of WTAP regulates m⁶A modification of interferon-stimulated genes

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Abstract

N⁶-methyladenosine (m⁶A) is the most prevalent modification of mRNA which controls diverse physiological processes. Although m⁶A modification is reported to regulate type I interferon (IFN) responses by targeting the mRNA of IFN-β and the interferon stimulated genes (ISGs), the detailed mechanism of how m⁶A methyltransferase complex (MTC) responses quickly to conduct the modification on nascent mRNA co-transcriptionally during IFN-β stimulation remains largely unclear. Here, we demonstrate that WTAP, the adaptor protein of m⁶A MTC, goes through dephosphorylation regulated phase transition from aggregates to liquid droplets under IFN-β stimulation. Phase transition of WTAP mediates the m⁶A modification of a subset of ISGs mRNA to restrict their expression. In mechanism, we found that formation of aggregates prevents WTAP from binding on the promoter region of ISGs or conducting m⁶A modification on mRNA in untreated cells. while IFN-β induced WTAP droplets interacts with nucleus-translocated transcriptional factor STAT1 and recruits MTC on the promoter region of ISGs, directing the co-transcriptional m⁶A modification on ISGs mRNA. Collectively, our findings reveal a novel regulatory role of WTAP phase transition under viral infection to orchestrate dynamic m⁶A modification with the cooperation of transcriptional factors and MTC, and precisely manipulate signaling pathway.

eLife assessment

This **important** study demonstrates that interferon beta stimulation induces WTAP transition from aggregates to liquid droplets, coordinating m6A modification of a subset of mRNAs that encode interferon-stimulated genes and restricting their expression. The evidence presented is **solid**, supported by microscopy, immunoprecipitations, m6A sequencing, and ChIP, to show that WTAP phosphorylation controls phase transition and its interaction with STAT1 and the methyltransferase complex.

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Introduction

Innate immunity is the first defense line against invading pathogens, mainly through the induction of type I interferon (IFN) and IFN-stimulated genes (ISGs) (Goubau, Deddouche, & Reis e Sousa, 2013). Once the production of type I IFN is initiated, the signal should be tightly restricted to maintain the homeostasis of immune responses (Henault et al., 2016; Schneider, Chevillotte, & Rice, 2014). In the past few years, a large number of studies have reported the post-translational modification of signaling molecules that regulate type I IFN signaling, such as phosphorylation (Uddin et al., 2002; Wen Z, Zhong Z, & Jr., 1995), ubiquitination (Cui et al., 2014; Qin et al., 2016) and methylation (Mowen et al., 2001). However, whether post-transcriptional modification of mRNA affects type I IFN responses remains largely unclear.

Methylation at the N^6 position of adenosine (m^6A) is the most pervasive post-transcriptional modification of mRNA. Deposition of m^6A is catalyzed by methyltransferase complex (MTC), including a key adaptor protein Wilms' tumor-associated protein (WTAP) and key catalytic proteins methyltransferase-like 3 (METTL3) and methyltransferase-like 14 (METTL14) (Schapira, 2016). The m^6A -modified RNAs are recognized by YT521B homology (YTH) domain family proteins, including YTHDC1, YTHDC2, YTHDF1, YTHDF2 and YTHDF3. m^6A modification are found to raise different cellular processes (Knuckles & Bühler, 2018; Meyer & Jaffrey, 2017), such as differentiation of stem cells (Geula et al., 2015), circadian clock (Zhong et al., 2018), splicing (Z. Zhang et al., 2010), translation (X. Wang et al., 2015) and destabilization of mRNA (X. Wang et al., 2014). Meanwhile, m^6A modification can be reversed by demethylase fat mass and obesity-associated gene (FTO) and AlkB homolog 5 (ALKBH5) (Jia et al., 2011; Zheng et al., 2013). The co-transcriptional m^6A modification is critical for new born mRNA produced by specific stimulation to participate in further physiological activities. m^6A modification of *IFN- β* mRNA was found to repress antiviral responses under infection of various viruses (Rubio, Depledge, Bianco, Thompson, & Mohr, 2018; Winkler et al., 2019), while others reported that m^6A deposition on different subsets of ISGs mRNA resulted in different effects, like promoting the translation of *IFITM1* mRNA (McFadden et al., 2021) and stabilizing the mRNA of *STAT1* and *IRF1* (L. Wang et al., 2020). Since the precise and timely m^6A modification on mRNA of antiviral genes plays a significant role in finetuning the antiviral responses, the function of non-catalytic protein WTAP and the detailed mechanism of how MTC responses to IFN- β stimulation need to be clarified.

Here, we observed the phase transition from aggregates to liquid droplets of WTAP driven by IFN- β -induced dephosphorylation. Based on the results from WTAP mutation assay, we revealed that under IFN- β stimulation, liquid-phase of WTAP cooperated with transcriptional factor STAT1 to recruit MTC on promoter region of ISGs, mediating the m^6A modification on new born ISGs mRNA

and the expression of a subset of ISGs, thus affecting the antiviral type I IFN responses. Our study sheds light on the mechanism that phase transition of WTAP bridges transcriptional factors and MTC to finetune effective m⁶A deposition on specific mRNAs.

Results

WTAP undergoes phase separation during IFN- β stimulation

To explore the detailed mechanism of WTAP-directed m⁶A modification on ISGs mRNA, we first investigated the expression of WTAP during virus infection. we infected cells by RNA virus VSV or DNA virus HSV-1 and conducted the immunofluorescence experiments. Surprisingly, we found WTAP clustered and formed condensate-like pattern in cells during virus infection (**Figure 1A** [↗](#) and **Figure 1** [↗](#)-figure supplement 1A). We then stimulated cells with IFN- β , and found the increasing number of WTAP condensates formation under IFN- β treatment (**Figure 1B** [↗](#)). Previous studies reported that some components in the methyltransferase complex such as METTL3 or YTHDFs performed liquid-liquid phase separation (LLPS) during m⁶A modification process (Han et al., 2022 [↗](#); J. Li et al., 2022 [↗](#)), we suspected whether the IFN- β -mediated WTAP condensates indicating the phase separation of WTAP.

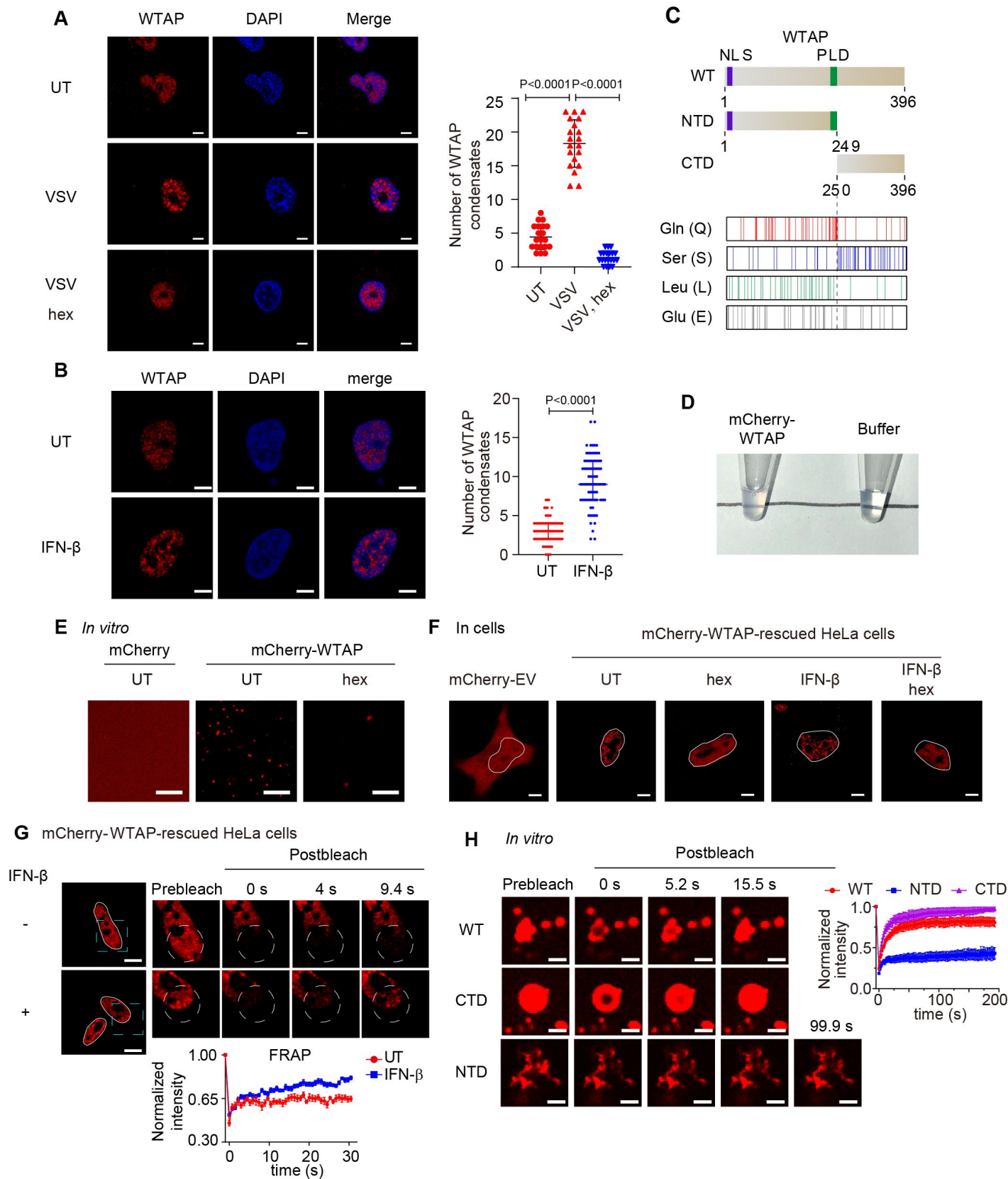


Figure 1

WTAP goes through phase transition with IFN- β stimulation.

(A) THP-1-derived macrophages were infected with VSV (m.o.i. = 0.1) for 24 hours together with or without 5% 1,6-hexanediol (hex) and 10 μ g/mL digitonin for 2 hours or left untreated (UT). WTAP (red) were imaged using confocal microscope. from n = 20 cells through ImageJ. Scale bars indicated 5 μ m.

(B) HeLa cells were placed on the dishes and stimulated with or without 10 ng/mL IFN- β for 1 hour at 37 °C. After stimulation, endogenous WTAP was stained and imaged using confocal microscopy. The number of WTAP condensates that diameter over 0.4 μ m of n = 80 cells were counted while medium with interquartile range was shown. Scale bars indicated 5 μ m.

(C) Domain structures (top) and distribution of amino acids (bottom) of WTAP protein. WT, wild type. NTD, N-terminal domain. CTD, C-terminal domain. NLS, nuclear localization signal. PLD, prion-like domain.

(D) Tubes containing physiological buffer with or without recombinant mCherry- WTAP (10 μ M) were compared, in which recombinant mCherry-WTAP underwent phase separation.

(E) Foci formation of recombinant mCherry-WTAP (10 μ M) with or without 10% 1,6-hexanediol (hex) was observed through confocal microscopy. Scale bars indicated 5 μ m.

(F) Phase separation of mCherry-WTAP in mCherry-WTAP-rescued HeLa cells treated with or without 5% 1,6-hexanediol (hex) and 20 μ g/mL digitonin were observed through confocal microscopy. Representative images of n=10 cells were shown. Scale bars indicated 10 μ m.

(G) mCherry-WTAP-rescued HeLa cells were placed on dishes and treated with or without 10 ng/mL IFN- β for 1 hour at 37 °C. After stimulation, bleaching of the WTAP foci was performed and quantification of FRAP of mCherry-WTAP aggregates was analyzed. The start time point of recovery after photobleaching was defined as 0 second. Representative images of n=10 cells were shown. Scale bars indicated 5 μ m.

(H) Recombinant mCherry-WTAP WT, NTD and CTD (40 μ M) were mixed with the physiological buffer and placed on the dishes at 37°C (Prebleach). After incubation, bleaching was performed and quantification of fluorescence recovery after photobleaching (FRAP) of recombinant mCherry-WTAP WT, NTD and CTD were analyzed. Representative images of n=6 condensates were shown, and the normalized intensity was measured and analyzed. The start time point of recovery after photobleaching was defined as 0 second. Scale bars indicated 2 μ m.

All error bars, mean values $\pm$ SD, P-values were determined by unpaired two-tailed Student's *t*-test of n = 20 cells in (A) and n=80 cells in (B). For (A-B and D-H), similar results were obtained for three independent biological experiments.

We predicted the prion-like domain (PLD) within WTAP using PLAAC websites (<http://plaac.wi.mit.edu/>) and found a fragment (234-249 amino acids) was predicted as PLD of WTAP, giving a clue that WTAP may have the ability to go through phase separation (**Figure 1C** and **Figure 1-figure supplement 1B**). To directly prove the phase separation properties of WTAP *in vitro*, we purified the recombinant mCherry- WTAP protein, mixed recombinant mCherry-WTAP with physiological buffer and found that the buffer became turbid (**Figure 1D**). Observation under microscope showed that mCherry-WTAP automatically formed liquid-like condensates which could be reversed by 1,6-hexanediol (hex), an inhibitor of phase separation (**Figure 1E**). Phase diagram of WTAP phase separation is also established under different concentration of potassium

chloride and recombinant mCherry-WTAP (**Figure 1** [↗](#)-figure supplement 1C). Fusion of condensates were observed and larger condensates were formed *in vitro* (**Figure 1** [↗](#)-figure supplement 1D). These data together confirmed that WTAP underwent phase separation *in vitro*.

To further explore the protein properties, we detected the phase separation behavior of WTAP in cells. Gene expressing mCherry-WTAP was introduced in WTAP-KO HeLa cells to generate mCherry-WTAP-rescued HeLa cells and treat with or without IFN- β stimulation. Results showed a few WTAP condensates occurred in untreated cells and increasing number of WTAP condensates under IFN- β stimulation. After hex treatment, WTAP condensates in both untreated cells or IFN- β treated cells were disrupted to a dispersed pattern (**Figure 1F** [↗](#)). Similarly, hex treatment in virus-infected cells also disrupted the formation of WTAP condensates (**Figure 1A** [↗](#) and Figure 1-figure supplement 1A). We next checked the mobility of WTAP condensates by fluorescence recovery after photobleaching (FRAP) experiments. Intriguingly, WTAP showed less recovery intensity in untreated cells, but showed higher mobility with quick recovery intensity in IFN- β treated cells (**Figure 1G** [↗](#)). These results implied that WTAP also underwent phase separation in cells, and IFN- β mediated the phase transition of WTAP to form the liquid-like droplets physiologically.

In order to figure out the driving force that mediates the phase separation of WTAP, we first detected the function of predicted PLD domain in WTAP. However, deletion of PLD failed to abolish the phase separation properties of WTAP (Figure 1-figure supplement 2A), showing that PLD-PLD interaction is not necessary for WTAP phase separation. Since previous studies have uncovered that protein phase separation could be controlled by interaction between specific region through multiple factors (Murthy et al., 2019 [↗](#); J. Wang et al., 2018 [↗](#)), including electrostatic interaction (Boyko, Qi, Chen, Surewicz, & Surewicz, 2019 [↗](#)), hydrophobic contacts (Reichheld, Muiznieks, Keeley, & Sharpe, 2017 [↗](#)) or hydrogen bonds (Gabryelczyk et al., 2019 [↗](#)), we next tried to figure out such phase separation-driven region of WTAP. We analyzed the net charge per region and hydropathy of WTAP, but no specific domains with opposite charge or high level of hydrophobicity was found (Figure 1-figure supplement 2B-C), indicating that electrostatic interaction and hydrophobic contacts might not be the key driving force for WTAP phase separation. We then analyzed the abundance of amino acids and found that Serine (Ser), Glutamine (Gln), Glutamate (Glu) and Leucine (Leu) were significantly higher than other amino acids within WTAP (Figure 1-figure supplement 2D). Gln and Leu were mainly located in the N-terminal domain (NTD) while Ser was mostly enriched in the C-terminal domain (CTD), implying the different properties of NTD and CTD (**Figure 1C** [↗](#)). Previous researches reported that hydrogen bond between serine side chain mediated the intermolecular interaction and LLPS of FUS. In addition, abnormal glutamine-repeat resulted in the formation of β -sheet and solid/gel-like aggregates with lower mobility (Perutz, 1996 [↗](#); Tanaka, Morishima, Akagi, Hashikawa, & Nukina, 2001 [↗](#)), which were related to neurodegenerative diseases like Huntington's disease (Gourfinkel-An et al., 1997 [↗](#)). Therefore, we wondered whether glutamine-rich NTD and serine-rich CTD of WTAP induced the distinct phase separation. As predicted, we found full-length WTAP and CTD WTAP formed liquid droplets, while NTD of WTAP clustered and formed aggregates with lower mobility, which was confirmed through FRAP experiments (**Figure 1H** [↗](#) and Figure 1-figure supplement 2E). Intriguingly, we further observed that full-length WTAP transitioned from droplets to aggregates after incubation for several minutes (Figure 1-figure supplement 2F), validating the different phase separation potential of NTD and CTD of WTAP.

IFN- β -mediated dephosphorylation induces the phase transition of WTAP

Since the serine-rich CTD played the key role in liquid droplets formation of WTAP, we wondered whether IFN- β -mediated WTAP phase transition through serine-phosphorylation by electrostatic repulsion among negatively charged phosphate. We detected the phosphorylation level of WTAP through IP assay and found that phosphorylation of WTAP was significantly decreased under IFN- β stimulation (**Figure 2A** [↗](#)). Through IP assays, we aimed to identify the phosphatases that

dephosphorylate WTAP, thus checked the interaction between WTAP and family of protein phosphatases (PPPs), and found that PPP4 presented the strongest interaction with WTAP (Figure 2-figure supplement 1A). IFN- β stimulation promoted the interaction between WTAP and PPP4 (**Figure 2B** [↗](#) and Figure 2-figure supplement 1B). Consistently, knockdown of PPP4 enhanced the phosphorylation level of WTAP in IFN- β treated cells (**Figure 2C** [↗](#)), indicating that IFN- β regulated dephosphorylation of WTAP through PPP4.

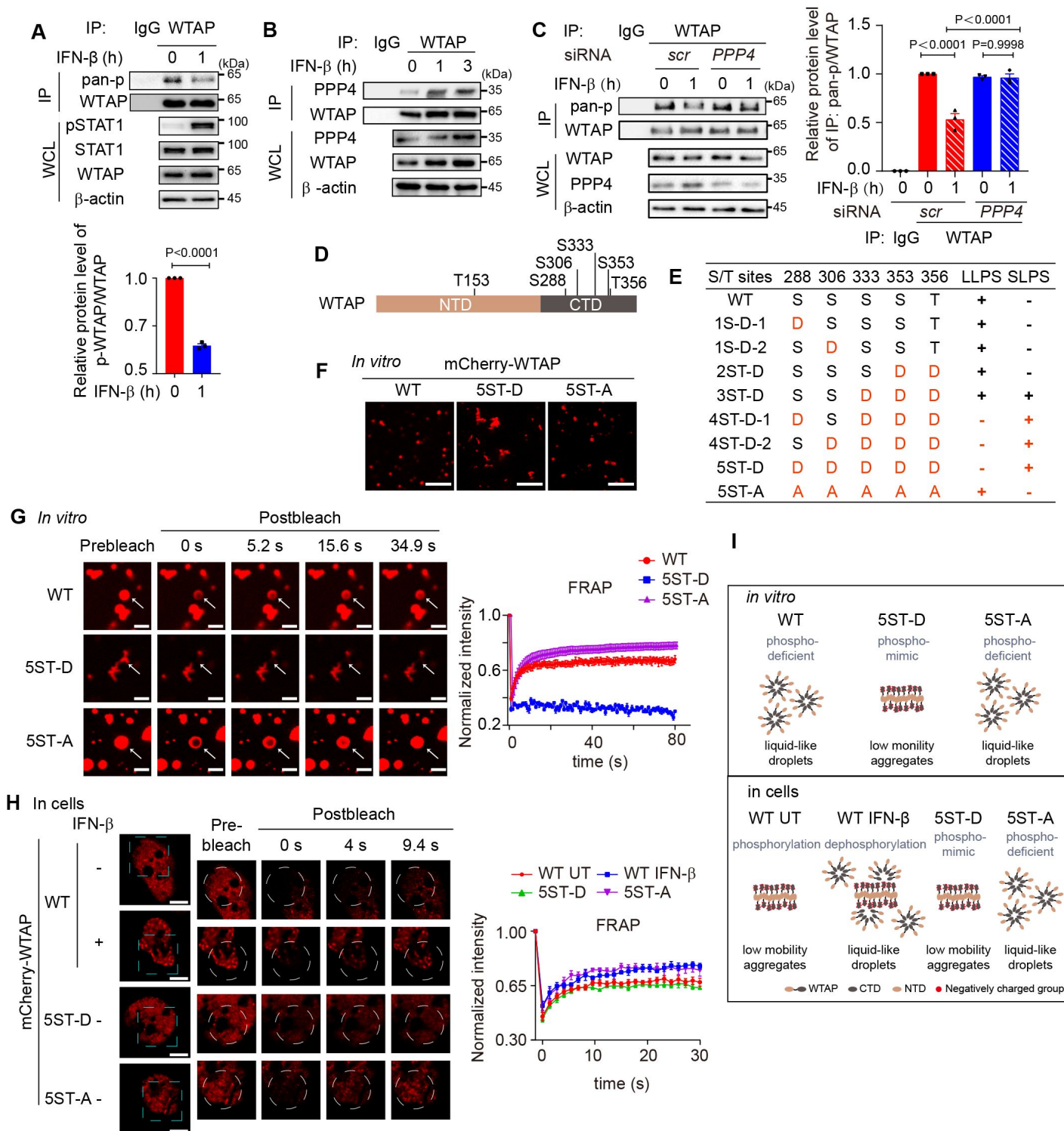


Figure 2.

IFN- β -mediated dephosphorylation of WTAP induces its phase transition.

(A) THP-1-derived macrophages were treated with 10 ng/mL IFN- β for 1 hour or left untreated. Whole cell lysates (WCL) were collected and immunoprecipitation experiment using anti-WTAP antibody or rabbit IgG was performed, followed with immunoblot using anti-phosphoserine/threonine/tyrosine antibody (pan-p). Relative protein level was shown.

(B) THP-1-derived macrophages were treated with 10 ng/mL IFN- β for 1 hour or left untreated. Whole cell lysates (WCL) were collected and immunoprecipitation experiment using anti-WTAP antibody or rabbit IgG was performed, followed with immunoblot.

(C) THP-1-derived macrophages were transfected with scramble (scr) or PPP4- targeted siRNA and stimulated with or without 10 ng/mL IFN- β for 1 hour. WCL were collected and IP experiment with anti-WTAP antibody or rabbit IgG was performed, followed with immunoblot using anti-phosphoserine/threonine/tyrosine antibody (pan-p). Relative protein level was measured and analyzed.

(D) mCherry-WTAP was purified from HEK 293T cells expressing mCherry-WTAP using anti-WTAP antibody and analyzed by mass spectrometry (MS) for the phosphorylation. Six phosphorylated residues were identified. Data of MS assay for five phosphorylated sites within CTD were shown in Figure 2-figure supplement 1C. (E-F) Recombinant mCherry-WTAP mutants with indicated Ser/Thr to Asp mutation were listed in (D), while representative images of the phase-separated mCherry- WTAP WT, 5ST-D and 5ST-A mutants were shown in (E). Images of phase separation of other mCherry-WTAP mutants were shown in Figure 2-figure supplement 1D. Scale bars indicated 5 μ m.

(G) Recombinant mCherry-WTAP WT (10 μ M), 5ST-A mutant (10 μ M) or 5ST-D mutant (10 μ M) was mixed with the physiological buffer and placed on the dishes at 37°C (Prebleach). After incubation, bleaching was performed and quantification of FRAP of recombinant mCherry-WTAP WT and 5ST-D mutant were analyzed. Representative images of n=5 condensates were shown and the normalized intensity was measured and analyzed. The start time point of recovery after photobleaching was defined as 0 second. Arrow indicated the FRAP area while scale bars indicated 2 μ m.

(H) HeLa cells expressing mCherry-WTAP WT, 5ST-D or 5ST-A mutants were placed on the dishes and stimulated with or without 10 ng/mL IFN- β for 1 hour at 37°C. After seeding, bleaching of the WTAP foci was performed and quantification of FRAP of mCherry-WTAP aggregates was analyzed. Representative images of n = 10 cells were shown, and the normalized intensity was measured and analyzed. The start time point of recovery after photobleaching was defined as 0 second. Representative images of n = 10 cells were shown. Scale bars indicated 5 μ m.

(I) Schematic figure of phase separation of WTAP WT, 5ST-D and 5ST-A *in vitro* (above) and WTAP WT with IFN- β stimulation or left untreated (UT), 5ST-D and 5ST-A in cells (below).

All error bars, mean values $\pm$ SD, P-values were determined by unpaired two-tailed Student's t-test of n = 3 independent biological experiments in (A and C). For (A-C and E-H), similar results were obtained for three independent biological experiments.

To validate the effects of phosphorylation on the phase transition of WTAP, we first confirming the phosphorylation site of WTAP by mass spectrometry (MS) assays. MS results identified six phosphorylated serine/threonine sites within WTAP, while five of six phosphorylation sites were located in CTD of WTAP (**Figure 2D** [↗](#) and Figure 2-figure supplement 1C). We then constructed a series of phosphorylation mimic, serine/threonine to aspartate mutants (S/T-D) or phosphorylation-deficient, serine/threonine to alanine mutant (S/T-A) of WTAP, and observed the

phase separation of WTAP mutants *in vitro*. We found that multiple sites ($n > 2$) with phosphorylation-mimic mutations of WTAP promoted the aggregation, while 5ST-A mutant formed similar liquid droplets as wild type (WT) WTAP (which was unphosphorylated due to the absence of kinase of WTAP in *E. coli*) (**Figure 2E-F** [↗](#) and Figure 2-figure supplement 1D). We next conducted FRAP experiments and found that mobility of WTAP wild type (WT) as well as 5ST-A was significantly higher than that of its 5ST-D mutant *in vitro* (**Figure 2G** [↗](#)). However, hyperphosphorylated WTAP WT in untreated cells showed lower mobility as 5ST-D mutant compared to phosphorylation-deficient 5ST-A mutant, while IFN- β -induced dephosphorylation promoted the recovery efficiency of WTAP WT as higher as 5ST-A mutants of WTAP (**Figure 2H** [↗](#)). Additionally, we found liquid droplets of WTAP after IFN- β stimulation was blunted in *PPP4* knocked-down cells (Figure 2-figure supplement 1E-F), revealed that IFN- β triggered *PPP4*-mediated dephosphorylation of WTAP to control the phase transition of WTAP. Taken together, we concluded that hyperphosphorylation of WTAP tended to form aggregates with low mobility due to electrostatically repulsed between phosphorylated serine sites, while hypophosphorylation of WTAP promotes the mobility and forms liquid droplets. IFN- β -mediated dephosphorylation promotes the transition from aggregates to liquid- phase of WTAP both *in vitro* and in cells (**Figure 2I** [↗](#)).

Phase transition of WTAP directs m⁶A modification on ISGs mRNA to regulate ISGs expression

Next, we wanted to figure out how IFN- β -mediated WTAP phase transition functions in ISGs mRNA. Previous reports indicated that m⁶A modification affected antiviral activity through enhancing stability and translation of ISGs mRNA (McFadden et al., 2021 [↗](#); L. Wang et al., 2020 [↗](#)). To verified the function of WTAP on ISGs mRNA m⁶A modification, we used CRISPR-Cas9 system to construct *WTAP*^{sgRNA} THP-1 cells, and performed the RNA sequencing (RNA-Seq) followed with bioinformatic analysis, and uncovered that 1440 and 1689 genes were up-regulated in response to IFN- β stimulation for 6 and 12 hours respectively (**Figure 3A** [↗](#) and Figure 3-figure supplement 1A-B), in which expression of 441 ISGs were enhanced in *WTAP*^{sgRNA} cells, and they were mainly enriched in various aspects of immune responses, including *IFIT1*, *IFIT2*, *OAS1* and *OAS2* (**Figure 3B-C** [↗](#)). We also found that WTAP showed no effect on the phosphorylation level or protein level of STAT1, STAT2 under IFN- β stimulation (**Figure 3** [↗](#)-figure supplement 1C-D), suggesting that WTAP regulated ISGs expression through other aspect but not the changes on transcriptional factors.

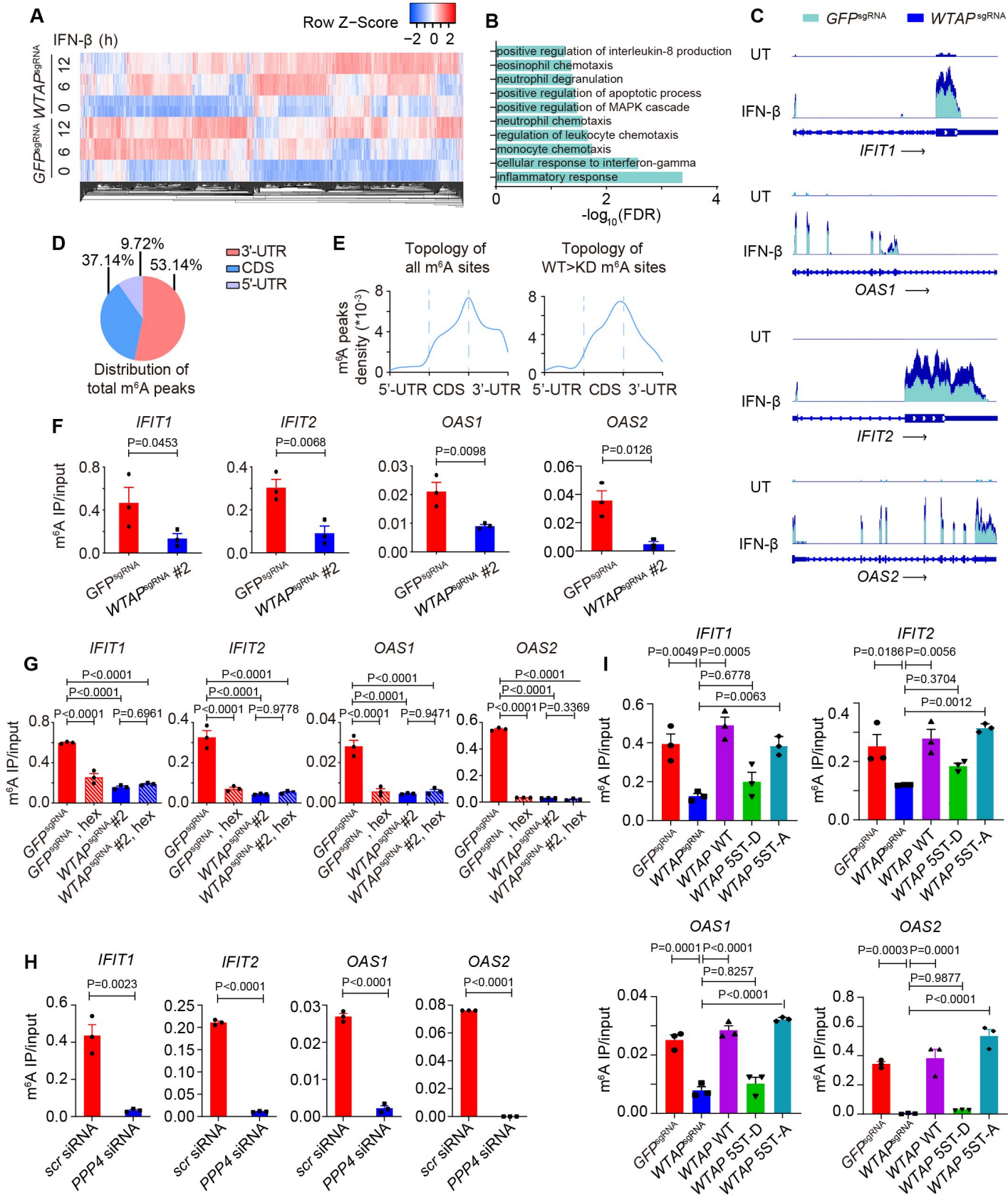


Figure 3.

WTAP phase transition mediates m⁶A modification to regulate ISG expression.

- (A) Transcriptome sequencing analysis of control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages stimulated with 10 ng/mL IFN- β for 0, 6, 12 hours. The count-per-million (CPM) value of the ISGs was drawn by Heatmapper and clustered using Centroid Linkage approach.
- (B) Gene ontology analysis for the WTAP down-regulated ISGs.
- (C) Transcriptome sequencing analysis of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* in control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages stimulated with 10 ng/mL IFN- β for 4 hours or left untreated (UT).
- (D-E) Distribution (D) and topology (E) analysis (5'-UTR, CDS and 3'-UTR) of total m⁶A sites and WTAP-dependent m⁶A sites.
- (F) Control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages were treated with 10 ng/mL IFN- β for 4 hours. MeRIP-qPCR assay of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).
- (G) Control (*GFP^{sgRNA}*) and *WTAP^{sgRNA}* #2 THP-1-derived macrophages were treated with 10 ng/mL IFN- β together with or without 5% 1,6-hexanediol (hex) and 20 μ g/mL digitonin for 2 hours or left untreated. After stimulation, medium was replaced by fresh 1640 medium while total RNA was collected 2 hours later. MeRIP-qPCR analysis of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A- modified mRNA and input were shown (m⁶A IP/input).
- (H) THP-1-derived macrophages transfected with scramble (*scr*) siRNA or *PPP4* siRNA were treated with 10 ng/mL IFN- β for 4 hours. MeRIP-qPCR assay of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).
- (I) Control (*GFP^{sgRNA}*) or *WTAP^{sgRNA}* #2 THP-1-derived macrophages with or without expression of Flag-tagged WTAP WT, and its 5ST-D or 5ST-A mutants were treated with 10 ng/mL IFN- β for 4 hours. MeRIP-qPCR assay of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* was performed and ratios between m⁶A-modified mRNA and input were shown (m⁶A IP/input).

All error bars, mean values $\pm$ SEM, P-values were determined by unpaired two-tailed Student's *t*-test of *n*=3 independent biological experiments in (F-I).

We then collected the mRNA of control and *WTAP^{sgRNA}* cells stimulated with or without IFN- β and performed m⁶A methylated RNA immunoprecipitation followed by deep sequencing (MeRIP-Seq). Consensus m⁶A modified core motifs were enriched in our samples and the majority of WTAP-dependent m⁶A modification peaks were located in CDS and 3'-UTR (Figure 3D-E and Figure 3-figure supplement 1E-F). ISGs that was m⁶A modified in control cells but diminished in *WTAP^{sgRNA}* cells were clustered in the IFN-associated pathway through MeRIP-seq assays (Figure 3-figure supplement 1G). By MeRIP- quantitative real time-polymerase chain reaction (qPCR) assays, we analyzed and confirmed the m⁶A modification level of WTAP-downregulated ISGs and found that deficiency of WTAP resulted in decrease of m⁶A level on ISGs, including *IFIT1*, *IFIT2*, *OAS1* and *OAS2* (Figure 3F). m⁶A deposition was reported to affect various aspects of RNA, especially for the decay of mRNA (Oerum, Meynier, Catala, & Tisne, 2021; Shi et al., 2017; X. Wang et al., 2014). We then checked the expression and mRNA stabilization of these ISGs by qPCR, and found that lack of m⁶A modification in *WTAP^{sgRNA}* cells led to the ISGs mRNA stabilization, thus up-regulated the ISGs expression (Figure 3-figure supplement 1H-I).

To further delineate the function of WTAP phase separation on m⁶A modification, we treated cells with hex to disrupt the phase separation of WTAP, and detected the m⁶A deposition in cells with virus infection or IFN-β treatment. We found that virus infection or IFN-β stimulation induced m⁶A modified ISGs mRNA, and their m⁶A modification was blocked by hex (**Figure 3G** and Figure 3-figure supplement 2A-B). Additionally, IFN-β-stimulated m⁶A modification enrichment on ISGs mRNA were inhibited with hex treatment in control cells but not *WTAP*^{sgRNA} cells (**Figure 3G**). Increased ISGs mRNA stability was also observed with hex treatment in control cells but not *WTAP*^{sgRNA} cells (Figure 3-figure supplement 2C), suggesting the essential role of WTAP phase separation in ISGs mRNA m⁶A deposition.

Next, we aimed to uncover the distinct ability to direct m⁶A modification of ISGs between aggregates or liquid-phase separated WTAP. By knocking down PPP4, hyperphosphorylated aggregation of WTAP were maintained under IFN-β stimulation, and the m⁶A modification on ISGs mRNA were surprisingly found to be decreased (**Figure 3H**). Enhanced mRNA level was also detected with PPP4 deficient (Figure 3-figure supplement 2D). We re-introduced WTAP WT, or its 5ST-D or 5ST-A mutant into *WTAP*^{sgRNA} THP-1 cells to generate WTAP WT, WTAP 5ST-D or 5ST-A THP-1 cells respectively (Figure 3-figure supplement 2E). Our results showed that m⁶A modification level of ISGs in *WTAP*^{sgRNA} and WTAP 5ST-D mutant cells was significantly attenuated than that of control cells, WTAP WT cells and WTAP 5ST-A mutant cells (**Figure 3I**). Collectively, these results demonstrated that m⁶A modification of ISGs mRNA was mainly regulated by the liquid-phase separated WTAP that dephosphorylated by PPP4 under IFN-β stimulation.

Liquid-phase separated WTAP recruits MTC and STAT1 on

promoter region to direct the m⁶A modification of ISGs mRNA

Although m⁶A modification of ISGs have been reported previously, the detailed mechanism of how WTAP directed m⁶A modification on ISGs mRNA and the distinct mechanism between WTAP aggregates and liquid-phase separated WTAP were still unknown. By checking the expressions of WTAP and other m⁶A modification- associated molecules, we found they were not significantly affected by IFN-β (**Figure 4**-figure supplement 1A-B). Previous report showed that m⁶A modification of TGF-β downstream genes was directed by the interaction between transcriptional factors SMAD2/3 and WTAP (Bertero et al., 2018). Therefore, we wondered whether m⁶A modification of ISGs mRNA was directed by WTAP and IFN-β-activated transcriptional factors, including STAT1, STAT2 and IRF9,³ which recruited the m⁶A methyltransferases for further m⁶A modification. We predicted the occupancy of the promoters of WTAP-regulated ISGs (2000 bp upstream of genes were analyzed, while 500 bp upstream of genes were considered as the proximal part of promoter) through directing using AnimalTFDB 3.0 website (<http://bioinfo.life.hust.edu.cn/AnimalTFDB/>) (Hu et al., 2019). Most of the ISGs were occupied by STAT1 rather than STAT2 and IRF9 (**Figure 4A**), especially for the proximal part of promoter, pointing the possible role of STAT1 in WTAP-induced m⁶A modification. We then detected the interaction between WTAP, METTL3 and STAT1 through immunoprecipitation (IP) assay and found that the interaction between WTAP, METTL3 and STAT1 was enhanced under IFN-β stimulation (**Figure 4B**). Additionally, the interaction between STAT1 and METTL3 was abolished in *WTAP*^{sgRNA} cells (**Figure 4C**). RIP-qPCR assays showed that interaction between METTL3 and ISGs mRNA was disrupted in *WTAP*^{sgRNA} cells, further validating the involvement and importance of WTAP in METTL3-dependent m⁶A deposition, and verifying the formation of STAT1-WTAP-METTL3-ISGs mRNA complex under IFN-β stimulation (**Figure 4D**).

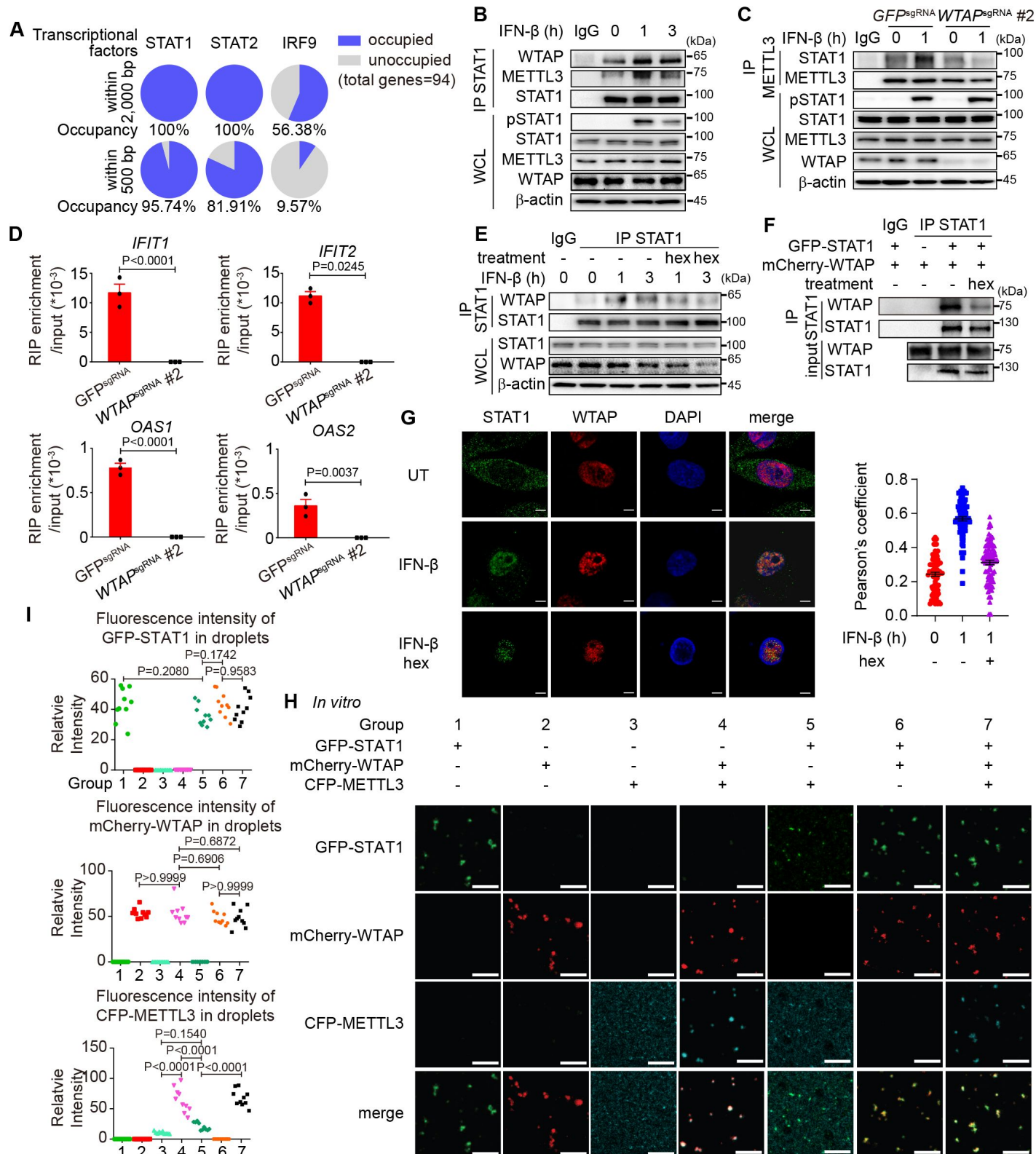


Figure 4.

Liquid-phase separated WTAP bridges STAT1 and m⁶A modification methyltransferase complex to direct m⁶A modification on ISGs mRNA.

- (A) Occupancy of the promoter of the WTAP down-regulated ISGs with lower m⁶A level in *WTAP*^{sgRNA} #2 cells (genes were identified in [Figure 3A](#) and both -2000-0 bp and -500-0 bp upstream of transcription start site were analyzed) by STAT1, STAT2 and IRF9 was predicted by AnimalTFDB 3.0.
- (B) THP-1-derived macrophages were treated with 10 ng/mL IFN-β for indicated time points. Whole cell lysates (WCL) were collected and immunoprecipitation (IP) experiment using anti-STAT1 antibody or rabbit IgG was performed, followed with immunoblot.
- (C) Control (*GFP*^{sgRNA}) and *WTAP*^{sgRNA} #2 THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 1 hour or left untreated. WCL were collected and IP experiment using anti-METTL3 antibody or rabbit IgG was performed, followed with immunoblot.
- (D) Control (*GFP*^{sgRNA}) and *WTAP*^{sgRNA} #2 THP-1-derived macrophages were treated with 10 ng/mL IFN-β for 2 hour or left untreated. RNA immunoprecipitation (RIP) experiments using anti-METTL3 antibody or rabbit IgG control were performed in control or *WTAP*^{sgRNA} #2 THP-1-derived macrophages treated with 10 ng/mL IFN-β for 2 hours. RNA of ISGs enriched by RIP was analyzed by qPCR assay, and the ratios between RIP-enriched RNA and input were shown (RIP enrichment/input).
- (E) THP-1-derived macrophages were treated with 10 ng/mL IFN-β for indicated time points together with or without 5% 1,6-hexanediol (hex) and 20 μg/mL digitonin. Whole cell lysates (WCL) were collected and IP experiment with anti-STAT1 antibody or rabbit IgG was performed, followed with immunoblot.
- (F) Recombinant GFP-STAT1 (10 μM) and mCherry-WTAP (10 μM) were incubated with or without 10% 1,6-hexanediol (hex). Immunoprecipitation (IP) experiment with anti-STAT1 antibody or rabbit IgG was performed, followed with immunoblot.
- (G) THP-1-derived macrophages were treated with 10 ng/mL IFN-β only or with 5% 1,6-hexanediol (hex) and 20 μg/mL digitonin for 1 hour or left UT. Interaction between STAT1 (green) and WTAP (red) were imaged using confocal microscope. Interaction between STAT1 and WTAP determined by Pearson's correlation coefficient was analyzed and calculated from n=80 cells through ImageJ. Scale bars indicated 5 μm.
- (H-I) Recombinant GFP-STAT1 (10 μM), mCherry-WTAP (10 μM) and CFP- METTL3 (10 μM) were incubated using physiological buffer at 37°C. After incubation, images were captured using confocal microscope (H). Relative fluorescence intensity of proteins in n=10 condensates were analyzed by ImageJ software (I).

All error bars, mean values ± SEM, P-values were determined by unpaired two-tailed Student's *t*-test of n = 3 independent biological experiments in (D). All error bars, mean values ± SD, P-values were determined by unpaired two-tailed Student's *t*-test of n = 3 independent biological experiments in (I). For (B-D, E-G and H), similar results were obtained for three independent biological experiments.

To determine the contribution of phase-separated WTAP, we used hex to inhibit WTAP phase separation and checked its interaction with STAT1. The interaction between WTAP, METTL3 and STAT1 was dramatically inhibited with hex treatment both in cells and *in vitro* ([Figure 4E-F](#) and [Figure 4-figure supplement 1C](#)). By immunofluorescence (IF) experiments, we found IFN-β promoted the interaction between WTAP, METTL3 and STAT1 in cells, which could be abolished by hex ([Figure 4G](#) and [Figure 4-figure supplement 1C](#)). In order to confirm the importance of

WTAP in bridging STAT1-METTL3 interaction, we purified the recombinant GFP- STAT1, mCherry-WTAP and CFP-METTL3 using *E. coli* expressing system and detected the condensation among these proteins *in vitro*. We observed that WTAP was able to form condensates with STAT1 or METTL3 alone while condensation of STAT1 and METTL3 was significantly diminished in the absence of WTAP (**Figure 4H-I** and Figure 4-figure supplement 1D). Consistent with the results in cells, the interaction between WTAP and STAT1 was blunted by hex (Figure 4-figure supplement 1E). Altogether, our data showed that phase separated WTAP functioned as the key adaptor protein bridging METTL3 and STAT1 to direct the m⁶A modification of ISGs mRNA.

We next tried to figure out how STAT1-MTC bound and mediated m⁶A modification on ISGs mRNA. By performing the chromosome immunoprecipitation (ChIP)-qPCR experiments, we found that WTAP interacted with promoter region of ISGs along with STAT1 under IFN- β stimulation (**Figure 5A-B**). Deficient of STAT1 abolished the binding affinity between WTAP and ISGs promoter region (**Figure 5C**), implying that nucleus-translocated transcriptional factor STAT1 directed WTAP and MTC to promoter region of ISGs to conduct m⁶A deposition process on mRNA. By knocking down PPP4 to block the phase transition to liquid droplets of WTAP, we observed the decreased interaction between WTAP and ISGs promoter region in *PPP4* knock-down cells, indicating the binding affinity between WTAP and ISGs promoter region might depend on PPP4-mediated phase transition of WTAP (**Figure 5C**). In addition, ChIP-qPCR experiments in WT, *WTAP*^{sgRNA}, WTAP WT, 5ST-D and 5ST-A THP-1 cells demonstrated that WTAP 5ST-D mutant showed significant reduced binding ability to ISGs promoter region after IFN- β stimulation compared to that of WTAP WT and WTAP 5ST-A cells (**Figure 5D**), which were consistent with the m⁶A modification results on ISGs mRNA (**Figure 3I**). These results further validated that phase transition of WTAP was important for the binding ability to ISGs promoter region and the m⁶A modification.

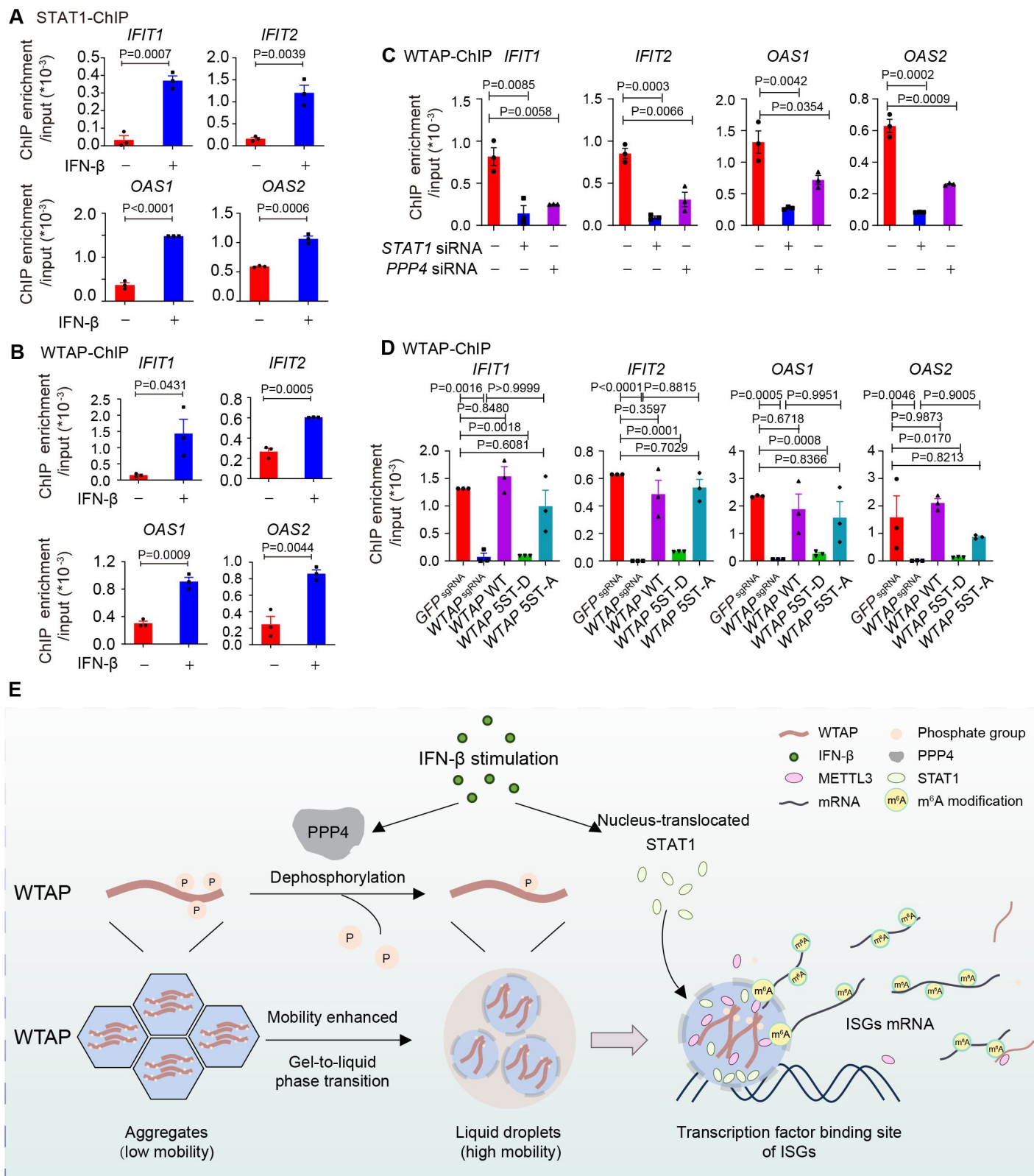


Figure 5.

Phase transition of WTAP mediates its interaction with ISGs promoter region to regulate m⁶A modification on ISGs mRNA.

(A) Chromatin immunoprecipitation (ChIP) experiments using anti-STAT1 antibody or rabbit IgG control were performed in THP-1-derived macrophages treated with 10 ng/mL IFN- β for 2 hours. Binding between the promoter region of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP was detected by qPCR assay and ratios between ChIP- enriched DNA and input were shown (ChIP enrichment/input).

(B) Chromatin immunoprecipitation (ChIP) experiments using anti-WTAP antibody or rabbit IgG control were performed in THP-1-derived macrophages treated with 10 ng/mL IFN- β for 2 hours or left untreated (UT). Binding between the promoters of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP were detected by qPCR assay, and the ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(C) Chromatin immunoprecipitation (ChIP) experiments using anti-WTAP antibody or rabbit IgG control were performed in THP-1-derived macrophages transfected with *scramble* (*scr*) siRNA, *STAT1* siRNA or *PPP4* siRNA treated with 10 ng/mL IFN- β for 2 hours. Binding between the promoter region of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP was detected by qPCR assay and ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(D) ChIP experiments using anti-WTAP antibody or rabbit IgG control were performed in control (*GFP^{sgRNA}*) or *WTAP^{sgRNA}* #2 THP-1-derived macrophages with or without expression of Flag-tagged wild type (WT) WTAP, and its SST-D or SST-A mutants treated with 10 ng/mL IFN- β for 2 hours. Binding between the promoter region of *IFIT1*, *IFIT2*, *OAS1* and *OAS2* with WTAP was detected by qPCR assay and ratios between ChIP-enriched DNA and input were shown (ChIP enrichment/input).

(E) Schematic figure of IFN- β -induced phase transition of WTAP regulates the m⁶A modification of ISGs mRNA.

All error bars, mean values $\pm$ SEM, P-values were determined by unpaired two-tailed Student's *t*-test of *n* = 3 independent biological experiments in (A-D).

Taken together, our findings revealed the driving force of aggregation and liquid droplets formation of WTAP, and demonstrated the distinct function between aggregates and liquid-phase separated WTAP. Based on our findings, we proposed a working model of WTAP phase transition in the regulation of m⁶A modification of ISGs mRNA under IFN- β stimulation. WTAP with high phosphorylation state in untreated cells tended to clustered together forming gel-like aggregates with lower mobility, and restricted the interaction between WTAP-dependent MTC and ISGs promoter region. After IFN- β stimulation, protein phosphatase PPP4 mediated the dephosphorylation of WTAP, resulting in the phase transition from aggregates to liquid-phase of WTAP. Liquid-phase of WTAP increased the mobility of WTAP droplets, and recruited METTL3 and nucleus-translocated STAT1 together to form STAT1-WTAP-METTL3 droplets. Leading by transcriptional factor STAT1, the droplets were able to directly bind with ISGs promoter region, and conducted the m⁶A modification of ISGs mRNA during the STAT1-mediated transcription process, thereby prompting diverse regulation of ISGs mRNA. Our work uncovered the function of WTAP phase transition physiologically through the cooperation between gene transcription and mRNA modification, and provided a full picture of multi-layer regulation of ISGs m⁶A modification and expression by the phase transition of WTAP (Figure 5E [↗](#)).

Discussion

m⁶A deposition is one of the most widespread post-transcriptional modifications, which is involved in various aspects of RNA metabolism to regulate different biological events, including antiviral response (Garcias Morales & Reyes, 2021). WTAP is a key component in MTC. Physiologically, WTAP works as a skeleton protein to recruit and stabilize other MTC components, such as METTL3 and METTL14, into nuclear speckles. Numerous studies have showed that WTAP was essential in embryo development, cell progression and differentiation, antiviral responses and many other physiological activities (Cho et al., 2021 [↗](#); Huang, Mo, Liao, Chen, & Zhang, 2022 [↗](#); X. Jiang et al., 2021 [↗](#)). Abnormal expression of WTAP was discovered in multiple diseases, including cancer or diabetes, to address dysfunctional m⁶A modification in specific genes (Chen et al., 2019 [↗](#); Huang et al., 2022 [↗](#); Z. X. Li et al., 2022 [↗](#); Yu et al., 2021 [↗](#)). However, the understanding of how WTAP builds up MTC, and how it reacts in response to specific stimulation is insufficient.

Phase separation is a well-known phenomenon associated with the formation of membraneless organelles, which concentrates the substrates and enzymes and accelerates biochemistry reactions (Alberti, Gladfelter, & Mittag, 2019 [↗](#)). Previous studies reported that METTL3 and YTHDFs underwent LLPS during the m⁶A deposition and recognition process (Han et al., 2022 [↗](#); J. Li et al., 2022 [↗](#)), while the m⁶A modification also promoted the LLPS of RNA-binding proteins, including m⁶A-mRNA-YTHDFs and m⁶A-eRNA/YTHDC1/BRD4 condensates (Gao et al., 2019 [↗](#); Lee et al., 2021 [↗](#); Ries et al., 2019 [↗](#); J. Wang et al., 2020 [↗](#)). But whether other molecules in MTC undergo phase separation and the driving force of their phase separation were still not clear. Here, we observed that WTAP also went through phase separation both *in vitro* and *in vivo*. We found WTAP mainly formed the aggregates with lower mobility in untreated cells, and transited to liquid droplets with IFN-β treatment, implying that phase separation participated in WTAP-dependent m⁶A deposition under IFN-β stimulation. By hex treatment and mutation assay to block phase transition of WTAP, we confirmed that low mobility of WTAP aggregates restricted the binding affinity to ISGs promoter region and inhibited the m⁶A deposition, while IFN-β-induced phase transition of WTAP resulted in the higher binding affinity and m⁶A modification level on ISGs mRNA. These results demonstrated the mechanism and the crucial role of WTAP phase separation and phase transition in MTC assembly and ISGs m⁶A deposition under IFN-β stimulation. Furthermore, the phase transition of WTAP suggested a potential mechanism that aggregation of WTAP might work as a backup storage, and the stimuli-induced phase transition was developed to perform the m⁶A modification timely and precisely in response to the requirement of cell development or abnormal situations. Thus, investigation of WTAP phase separation in different model could be conducted to explain the mechanism of m⁶A modification under different stimulation and explore possible therapeutic targets of multiple diseases.

Post-translational modifications are one of the most common ways to modulate characteristic of proteins (Luo, Wu, & Li, 2021 [↗](#)), like phosphorylation of serine and threonine (Markevich, Hoek, & Kholodenko, 2004 [↗](#)), methylation of arginine (Guccione & Richard, 2019 [↗](#)). Several reports indicated that phosphorylation of phase-separated proteins are associated with the solid/gel-like aggregates formation or solid/gel-to-liquid phase transition, participating in various physiological process such as the pathogenesis of tau protein aggregates or acute transcriptional response (Arendt, Stieler, & Holzer, 2016 [↗](#); Boyko et al., 2019 [↗](#); H. Zhang et al., 2022 [↗](#)), but whether the phosphorylation-controlled phase transition of proteins regulated other physiological process remains to be illustrated. In this study, we observed that phosphorylation of WTAP promoted the aggregates formation with lower mobility of WTAP, while IFN-β induced the dephosphorylation of WTAP by PPP4 at multiple serine sites, led to the gel-to-liquid phase transition of WTAP. Consistently, phosphomimic 5ST-D mutant of WTAP mainly aggregated as lower mobility form compared to phosphodeficient 5ST-A mutant of WTAP, which formed liquid droplets in cells,

indicating that phosphorylation level of WTAP balanced its phase transition. Therefore, our findings shed light on the mechanism of post-translational modification-regulated phase transition of WTAP and put forward an open question whether phosphorylation and other post-translational modifications function as the key strategy for global phase transition of proteins *in vivo*.

As a global modification in cells, m⁶A modification could be hindered by MTC along with the inhibition role of exon junction complex, led to the specificity of modification, mapping the m⁶A topology and maintain in stable state (He et al., 2023 [DOI](#); Uzonyi et al., 2023 [DOI](#)). More and more results suggesting that MTC could be recruited and enriched in different chromatin loci, including promoter or enhancer, to dynamically deposit m⁶A modification on nascent mRNA co-transcriptionally under certain circumstance or function in various physiological activities, implying the importance of interaction between MTC and chromatin architecture (Dou et al., 2023 [DOI](#); Sendinc & Shi, 2023 [DOI](#); Xu et al., 2022 [DOI](#)). In this work, we validated the IFN-β-induced STAT1-WTAP-METTL3 interaction guiding MTC to the promoter region of ISGs to direct the m⁶A modification on ISGs mRNA and regulate the mRNA decay. Similarly, Bertero *et al* found that TGF-β-activated transcriptional factor SMAD2/3 recruits MTC and specifically directed the m⁶A modification and degradation of a subset of downstream transcripts, which affects the early cell fate decision (Bertero et al., 2018 [DOI](#)). These studies uncovered the mechanism of phase transition-mediated transcriptional factor-MTC interaction directed m⁶A modification of downstream genes co-transcriptionally in maintaining the cellular homeostasis under specific stimulation, raised a new perspective on the cooperation of gene transcription and mRNA modification.

Taken together, our study drew a full picture of the phase transition behavior of WTAP and the function of STAT1-WTAP-MTC directed m⁶A modification on ISGs mRNA under IFN-β stimulation. Our findings unraveled a novel mechanism of finetuning m⁶A methylated mRNA profile under stimulation, and provided a possible therapeutic target in antiviral responses and many other diseases.

Materials and Methods

Cells

THP-1 cells obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China) were cultured in RPMI 1640 (Gibco) with 10% (vol/vol) fetal bovine serum (FBS), 1% penicillin-streptomycin (Gibco, 1:100) and 1% L-glutamine (Gibco) incubated in a 37°C chamber with 5% CO₂ (Thermo Fisher Scientific). Before stimulation, THP-1 cells were differentiated into macrophages (THP-1-derived macrophages) through treatment of 100 nM phorbol-12-myristate-13-acetate (PMA) (P8139, Sigma) for 16 hours. After PMA treatment, the macrophages were rested for 48 hours before stimulation. A549 and HeLa cells obtained from the Cell Bank of the Chinese Academy of Sciences (Shanghai, China) were cultured in DMEM medium (Corning) with 10% FBS and 1% L-glutamine (Gibco) incubated in a 37°C chamber with 5% CO₂.

Reagents and Antibodies

IFN-β (cat. #300-02BC) was purchased from PeproTech Inc. PMA (cat. #P1585), puromycin (cat. #P9620), Actinomycin D (cat. #SBR00013), 1,6-hexanediol (cat. #240117), digitonin (cat. #D141), DTT (cat. #3483-12-3), DAPI (cat. #D9542) were purchased from Sigma-Aldrich. Isopropyl-beta-D-thiogalactopyranoside (IPTG, cat. #CA413) and Superluciferase™ High-efficiency Transfection Reagent (cat. #11231804) were purchased from MIKX. NP-40 (cat. #P0013F) was purchased from Beyotime Biotechnology. The antibodies that used in this study were listed in *Table S1*.

Generation of knockout cell lines by CRISPR/Cas9

After THP-1 cells were seeded, medium was replaced by DMEM containing polybrene (8 µg/ml) (Sigma-Aldrich) lentiviral vector encoding Cas9 and small guide RNAs (sgRNA) for 48 hours. Cells were selected using puromycin (Sigma-Aldrich). The sequence of sgRNAs targeting indicated gene obtained from Sangon (Shanghai, China) were listed in *Table S2*:

siRNA Transfection

siRNA duplexes and scrambled siRNA were chemically synthesized by RIBOBIO (Guangzhou, China) and transfected into cells using LipoRNAiMAX reagent (Invitrogen) according to the manufacturer's instructions. The sequences of siRNAs were listed in *Table S2*:

Virus infection

VSV and HSV-1 were amplified and tittered on Vero E6 cell line. Virus titers were measured by means of 50% of the tissue culture's infectious dose (TCID₅₀). Virus were infected into indicated cell lines with indicated titers as shown in figure legends.

Immunoblot (IB) assays and immunoprecipitation (IP)

Relevant cells were stimulated as indicated, and whole cell lysate was obtained using low-salt lysis buffer (50 mM Hepes (pH 7.5), 150 mM NaCl, 1 mM EDTA, 1.5 mM MgCl₂, 10% glycerol, 1% Triton X-100, supplemented with protease inhibitor cocktail (cat. #A32965, Pierce) and phosSTOP Phosphatase Inhibitor Cocktail (cat. #4906837001, Roche)). Cell lysates were centrifuged at 4°C, 12000 g for 15 minutes and supernatants was boiled at 100 °C for 5 minutes with the 5×Loading Buffer (cat. #FD006, Hangzhou Fude Biological Technology Co, LTD.). Solution was resolved by SDS-PAGE, transferred to polyvinylidene fluoride (PVDF) membranes (cat. #1620177, Bio-Rad Laboratories, Inc.), blocked with 5% skim milk (cat. #232100, BD) for 1 hour and incubated with appropriate antibody. Immobilon Western HRP Substrate (cat. #WBKLS0500, Millipore) was used for protein detection with ChemiDoc MP System (Bio-Rad Laboratories, Inc.) and Image Lab version 6.0 (Bio-Rad software, California, USA). For immunoprecipitation, protein samples were incubated with protein A/G beads (cat. #20333, #20399, Pierce) or anti-Flag M2 affinity gel (cat. #A2220, Sigma-Aldrich) together with indicated antibody at 4°C overnight. Beads were washed five times with low-salt lysis buffer and boiled at 100 °C for 5 minutes with 2×SDS loading buffer, followed with immunoblot assays described above.

Mass spectrometry

HEK 293T cells were transfected with plasmids expressing mCherry-WTAP. 24 hours after transfection, whole cell lysate was obtained using low-salt lysis buffer and immunoprecipitated using protein A/G beads together with anti-WTAP antibody at 4°C overnight. Proteins were obtained from beads and analyzed by LC-MS/MS analysis. Details were shown in *SI Appendix, Extended Methods*.

RNA extraction, quantitative real-time polymerase chain reaction (qPCR) and RNA sequencing (RNA-Seq) assay

Relevant cells were treated as indicated and total RNA was extracted from cells with TRIzol reagent (cat. #15596026, Invitrogen), according to the manufacturer's instructions. RNA was then reverse-transcribed into cDNA using HiScript[®] III RT SuperMix for qPCR (+gDNA wiper) (cat. #R323-01, Vazyme). The level of indicated genes were determined by qPCR with 2×PolarSignal[™] SYBR Green mix Taq (cat. #MKG900-10, MIKX) using LightCycler[®] 480 System (Roche). The primers

used in qPCR were listed in Table S2. For RNA-Seq assay, total RNA was isolated from cells using TRIzol reagent, and sequencing was performed by Sangon Biotech (Shanghai, China). Details of RNA-Seq data analysis were shown in *SI Appendix, Extended Methods*.

Chromatin-immunoprecipitation (ChIP) assay

THP-1 cells were treated as indicated and DNA were enriched through ChIP assay referred to the Rockland Chromatin Immunoprecipitation assay protocol. Details were shown in *SI Appendix, Extended Methods*. After ChIP enrichment, DNA were analyzed through qPCR assay. The sequences of primers were listed in Table S2:

RNA-immunoprecipitation (RIP) followed with qPCR assay

THP-1 cells were treated as indicated and RNA were enriched through RIP assay referred to the previously reported protocol (Keene, Komisarow, & Friedersdorf, 2006 [\[3\]](#)). Details were shown in *SI Appendix, Extended Methods*. After RIP enrichment, RNA were analyzed through qPCR assay. The sequences of primers were listed in Table S2:

m⁶A methylated RNA immunoprecipitation (MeRIP) followed with qPCR assay and MeRIP-Seq assay

For MeRIP-qPCR, the procedures were referred to the protocol as manufacturer's instructions. Approximately 2×10^7 cells were seeded and stimulated by 10 ng/mL IFN- β for 6 hours. Approximately more than 50 μ g of total RNA was subjected to isolate poly (A) mRNA with poly-T oligo attached magnetic beads (cat. #61006, Invitrogen). Following purification, the poly(A) mRNA fractions is fragmented into ~ 100 -nt-long oligonucleotides using divalent cations under elevated temperature. Then the cleaved RNA fragments were subjected to incubation for 2h at 4°C with m⁶A-specific antibody (cat. #202003, Synaptic Systems) in IP buffer (50 mM Tris- HCl, 750 mM NaCl and 0.5% Igepal CA-630) supplemented with BSA (0.5 μ g/ μ L). The mixture was then incubated with protein-A beads and eluted with elution buffer (1 $\times$ IP buffer and 6.7mM m⁶A), while 10% of fragmented mRNA was kept as input. After immunoprecipitation, precipitates were washed by IP buffer for 5 times and RNA was isolated by adding TRIzol reagent. Purified m⁶A-containing mRNA fragments and untreated mRNA input control fragments are converted to final cDNA library in accordance with a strand-specific library preparation by dUTP method, and analyzed by qPCR assay. The primers used in qPCR were listed in Table S2.

For MeRIP-Seq, approximately 5×10^7 cells were seeded and stimulated by 10 ng/mL IFN- β for 4 hours. After stimulation, total RNA was extracted using TRIzol reagent. The total RNA quality and quantity were analysis of Bioanalyzer 2100 and RNA 6000 Nano LabChip Kit (Agilent, CA, USA) with RIN number >7.0, followed with MeRIP assay. And then we performed the paired-end 2 $\times$ 150 bp sequencing on an Illumina Novaseq 6000 platform at the LC-BIO Bio-tech ltd (Hangzhou, China) following the vendor's recommended protocol. Details of MeRIP-Seq data analysis were shown in *SI Appendix, Extended Methods*.

Immunofluorescence staining (IF) and imaging

THP-1-derived macrophages were differentiated and stimulated as described above. After stimulation, cells were fixed by in 4% paraformaldehyde (Meilunbio, China) for 15 minutes at room temperature, permeated by ice-cold methanol at -20 °C for 15 minutes and blocked using 6% goat serum (cat. #AR1009, Boster Biological) for 1 hour, and then incubated with indicated primary antibody at 4°C overnight, followed by incubation with indicated secondary antibody at room temperature for 1 hour. Phosphate buffer (PBS) was used to wash three times for 5 minutes between each step. After staining, Leica TCS-SP8 STED 3X confocal fluorescence microscope was used to acquire images. Images were captured by Leica TCS-SP8 STED 3X confocal fluorescence microscope and analyzed by Leica Application Suite Advanced Fluorescence (LAS AF, Version 4.2)

software, while deconvolution was conducted by Huygens Software (Version 23.04). The colocalization between proteins was quantified by ImageJ software (National Institutes of Health, Maryland, USA, Version 1.52) using Pearson's correlation coefficient, ranging from -1 to 1. A value of -1 represents perfect negative correlation, 0 means no correlation and 1 indicates perfect correlation.

Expression, purification, and interaction of mCherry-WTAP WT, 5ST-A/ST-D mutants, CFP-METTL3 and GFP-STAT1

Procedures of the expression of proteins in *E. coli* were referred to Hao Jiang et al. (H. Jiang et al., 2015 [DOI](#)). Recombinant mCherry-WTAP, CFP-METTL3 and GFP-STAT1 were cloned into pET-28A plasmid, while ST-D and 5ST-A mutants of WTAP were constructed by mutating the pET-28A-mCherry-WTAP plasmid using Muta-directTM Kit (Sbsgene, cat. #SDM-15). After transformed into *E. coli*, protein was extracted as shown in *SI Appendix, Extended Methods*.

For detection of interaction among mCherry-WTAP or its indicated mutants, CFP-METTL3 and GFP-STAT1, purified proteins were mixed using physiological buffer (20 mM Tris-HCl, pH 7.5, 150 mM KCl, 10% PEG8000) in the final concentration of 10 μ M and placed on the slides. Mixtures were incubated and imaged at 37°C in a live-cell-imaging chamber of Leica TCS-SP8 STED 3X confocal fluorescence microscope while images were analyzed by LAS AF software.

Phase separation assay and fluorescence recovery after photobleaching (FRAP)

Expression, purification, and interaction of mCherry-WTAP WT, 5ST-A/ST-D mutants, CFP-METTL3 and GFP-STAT1 were shown in *SI Appendix, Extended Methods*. For *in vitro* phase separation assay, recombinant mCherry-WTAP WT, NTD, CTD, Δ PLD, 5ST-D and 5ST-A mutants in indicated concentration were mixed with physiological buffer (20 mM Tris-HCl, pH 7.5, 150 mM KCl, 10% PEG8000). After incubation, turbid solution in the 200 μ L PCR tubes were directly observed. For further capture of fluorescence images, solution of recombinant mCherry-WTAP and physiological buffer were treated with or without hex at a final concentration of 10% and placed on dishes. Leica TCS-SP8 STED 3X confocal fluorescence microscope with live-cell-imaging chamber at 37°C was used to capture fluorescence images and the 3D images. For *in cells* separation assay, plasmids expression mCherry-WTAP WT, 5ST-D and 5ST-A were transfected into HeLa cells for 24 hours, and treated with or without 5% hex and 20 μ g/mL digitonin for 1 hour. Leica TCS-SP8 STED 3X confocal fluorescence microscope with live-cell-imaging chamber at 37°C was used to capture fluorescence images. For observing the fusion of condensates, recombinant mCherry-WTAP mixed with physiological buffer or HeLa cells transfected with mCherry-WTAP plasmids were incubated in the live-cell-imaging chamber at 37°C and. Images were captured by Leica TCS-SP8 STED 3X confocal fluorescence microscope, while deconvolution was conducted by Huygens software.

For *in vitro* FRAP experiments were performed on Leica TCS-SP8 STED 3X confocal fluorescence microscope. Recombinant mCherry-WTAP WT, NTD, CTD, 5ST-D and 5ST-A mutants in indicated concentration was mixed with physiological buffer on dishes. For *in cells* FRAP experiment, HeLa cells transfected with mCherry-WTAP WT, 5ST-D and 5ST-A mutants were treated as indicated. Samples were imaged with Leica TCS-SP8 STED 3X confocal fluorescence microscope with live-cell-imaging chamber at 37°C. Spots of mCherry-WTAP foci were photobleached with 100% laser power using 567-nm lasers followed with time-lapse images. Images were analyzed by LAS AF software. Fluorescence intensities of regions of interest (ROIs) were corrected by unbleached control regions and then normalized to pre-bleached intensities of the ROIs.

Statistical Analysis

The data of all quantitative experiments are represented as mean $\pm$ SEM of three independent experiments. Statistical analyses were performed with GraphPad Prism software version 8.0 (GraphPad software, California, USA) using two-way analysis of variance (ANOVA) followed by Tukey's test or unpaired two-tailed Student's *t*-test as described in the figure legends. P-value < 0.05 was considered as statistically significant of all statistical analyses.

Data availability

Raw data of RNA-Seq and MeRIP-Seq have been deposited in the Sequence Read Archive (SRA) database under accession code PRJNA680033 and PRJNA680974, respectively. The data that support this study are available within the article and its Supplementary Information files or available from the authors upon request.

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Additional information

Author Contributions

J.C. conceived the project. *J.C.*, *S.C.* and *J.Z.* designed the experiments. *S.C.*, *J.Z.* and *S.J.* expressed and purified the proteins using *E. coli* system while *S.C.*, *J.Z.* and *C.Z.* finished the rest of the experiments. *X.L.*, *J.R.*, *S.C.* and *J.Z.* analyzed the data of RNA-Seq and MeRIP-Seq. All authors discussed the results and commented on or prepared the manuscript.

Competing Interest

The authors declare no competing interests.

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Reviewer #1 (Public review):

Summary:

This study puts forth the model that under IFN- β stimulation, liquid-phase WTAP coordinates with the transcription factor STAT1 to recruit MTC to the promoter region of interferon-stimulated genes (ISGs), mediating the installation of m6A on newly synthesized ISG mRNAs. This model is supported by strong evidence that the phosphorylation state of WTAP, regulated by PPP4, is regulated by IFN- β stimulation, and that this results in interactions between

WTAP, the m6A methyltransferase complex, and STAT1, a transcription factor that mediates activation of ISGs. This was demonstrated via a combination of microscopy, immunoprecipitations, m6A sequencing, and ChIP. These experiments converge on a set of experiments that nicely demonstrate that IFN- β stimulation increases the interaction between WTAP, METTL3, and STAT1, that this interaction is lost with the knockdown of WTAP (even in the presence of IFN- β), and that this IFN- β stimulation also induces METTL3-ISG interactions.

Strengths:

The evidence for the IFN- β stimulated interaction between METTL3 and STAT1, mediated by WTAP, is quite strong. Removal of WTAP in this system seems to be sufficient to reduce these interactions and the concomitant m6A methylation of ISGs. The conclusion that the phosphorylation state of WTAP is important in this process is also quite well supported.

Weaknesses:

The evidence that the above mechanism is fundamentally driven by different phase-separated pools of WTAP (regulated by its phosphorylation state) is weaker. These experiments rely relatively heavily on the treatment of cells with 1,6-hexanediol, which has been shown to have some off-target effects on phosphatases and kinases (PMID 33814344). Given that the model invoked in this study depends on the phosphorylation (or lack thereof) of WTAP, this is a particularly relevant concern. Related to this point, it is also interesting (and potentially concerning for the proposed model) that the initial region of WTAP that was predicted to be disordered is in fact not the region that the authors demonstrate is important for the different phase-separated states. Taking all the data together, it is also not clear to me that one has to invoke phase separation in the proposed mechanism.

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Reviewer #2 (Public review):

In this study, Cai and colleagues investigate how one component of the m6A methyltransferase complex, the WTAP protein, responds to IFN β stimulation. They find that viral infection or IFN β stimulation induces the transition of WTAP from aggregates to liquid droplets through dephosphorylation by PPP4. This process affects the m6A modification levels of ISG mRNAs and modulates their stability. In addition, the WTAP droplets interact with the transcription factor STAT1 to recruit the methyltransferase complex to ISG promoters and enhance m6A modification during transcription. The investigation dives into a previously unexplored area of how viral infection or IFN β stimulation affects m6A modification on ISGs. The observation that WTAP undergoes a phase transition is significant in our understanding of the mechanisms underlying m6A's function in immunity. However, there are still key gaps that should be addressed to fully accept the model presented.

Major points:

- (1) More detailed analyses on the effects of WTAP sgRNA on the m6A modification of ISGs:
 - a. A comprehensive summary of the ISGs, including the percentage of ISGs that are m6A-modified.
 - b. The distribution of m6A modification across the ISGs.
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In addition, since the authors propose a novel mechanism where the interaction between phosphorylated STAT1 and WTAP directs the MTC to the promoter regions of ISGs to facilitate co-transcriptional m6A modification, it is critical to analyze whether the m6A modification distribution holds true in the data.

(2) Since a key part of the model includes the cytosol-localized STAT1 protein undergoing phosphorylation to translocate to the nucleus to mediate gene expression, the authors should focus on the interaction between phosphorylated STAT1 and WTAP in Figure 4, rather than the unphosphorylated STAT1. Only phosphorylated STAT1 localizes to the nucleus, so the presence of pSTAT1 in the immunoprecipitate is critical for establishing a functional link between STAT1 activation and its interaction with WTAP.

(3) The authors should include pSTAT1 ChIP-seq and WTAP ChIP-seq on IFN β -treated samples in Figure 5 to allow for a comprehensive and unbiased genomic analysis for comparing the overlaps of peaks from both ChIP-seq datasets. These results should further support their hypothesis that WTAP interacts with pSTAT1 to enhance m6A modifications on ISGs.

Minor points:

(1) Since IFN β is primarily known for modulating biological processes through gene transcription, it would be informative if the authors discussed the mechanism of how IFN β would induce the interaction between WTAP and PPP4.

(2) The authors should include mCherry alone controls in Figure 1D to demonstrate that mCherry does not contribute to the phase separation of WTAP. Does mCherry have or lack a PLD?

(3) The authors should clarify the immunoprecipitation assays in the methods. For example, the labeling in Figure 2A suggests that antibodies against WTAP and pan-p were used for two immunoprecipitations. Is that accurate?

(4) The authors should include overall m6A modification levels quantified of GFPsgRNA and WTAPsgRNA cells, either by mass spectrometry (preferably) or dot blot.

<https://doi.org/10.7554/eLife.100601.1.sa2>

Reviewer #3 (Public review):

Summary:

This study presents a valuable finding on the mechanism used by WTAP to modulate the IFN- β stimulation. It describes the phase transition of WTAP driven by IFN- β -induced dephosphorylation. The evidence supporting the claims of the authors is solid, although major analysis and controls would strengthen the impact of the findings. Additionally, more attention to the figure design and to the text would help the reader to understand the major findings.

Strength:

The key finding is the revelation that WTAP undergoes phase separation during virus infection or IFN- β treatment. The authors conducted a series of precise experiments to uncover the mechanism behind WTAP phase separation and identified the regulatory role of 5 phosphorylation sites. They also succeeded in pinpointing the phosphatase involved.

Weaknesses:

However, as the authors acknowledge, it is already widely known in the field that IFN and viral infection regulate m6A mRNAs and ISGs. Therefore, a more detailed discussion could help the reader interpret the obtained findings in light of previous research.

It is well-known that protein concentration drives phase separation events. Similarly, previous studies and some of the figures presented by the authors show an increase in WTAP expression upon IFN treatment. The authors do not discuss the contribution of WTAP expression levels to the phase separation event observed upon IFN treatment. Similarly, METTL3 and METTL14, as well as other proteins of the MTC are upregulated upon IFN treatment. How does the MTC protein concentration contribute to the observed phase separation event?

How is PP4 related to the IFN signaling cascade?

In general, it is very confusing to talk about WTAP KO as WTAPgRNA.

<https://doi.org/10.7554/eLife.100601.1.sa1>

Author response:

Public Reviews:

Reviewer #1 (Public review):

Summary:

This study puts forth the model that under IFN- β stimulation, liquid-phase WTAP coordinates with the transcription factor STAT1 to recruit MTC to the promoter region of interferon-stimulated genes (ISGs), mediating the installation of m6A on newly synthesized ISG mRNAs. This model is supported by strong evidence that the phosphorylation state of WTAP, regulated by PPP4, is regulated by IFN- β stimulation, and that this results in interactions between WTAP, the m6A methyltransferase complex, and STAT1, a transcription factor that mediates activation of ISGs. This was demonstrated via a combination of microscopy, immunoprecipitations, m6A sequencing, and ChIP. These experiments converge on a set of experiments that nicely demonstrate that IFN- β stimulation increases the interaction between WTAP, METTL3, and STAT1, that this interaction is lost with the knockdown of WTAP (even in the presence of IFN- β), and that this IFN- β stimulation also induces METTL3-ISG interactions.

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Related to this point, it is also interesting (and potentially concerning for the proposed model) that the initial region of WTAP that was predicted to be disordered is in fact not the region that the authors demonstrate is important for the different phase-separated

states. Taking all the data together, it is also not clear to me that one has to invoke phase separation in the proposed mechanism.

We are grateful for the Reviewer's positive comment and constructive feedback. In this article, we claim a novel and important mechanism that de-phosphorylation-driven solid to liquid phase transition of WTAP mediates its co-transcriptional m6A modification. We first observed that WTAP underwent phase transition during virus infection and IFN- β stimulation, and confirmed the phase transition driven force of WTAP through multiple experiments. Besides 1,6-hexanediol (1,6-hex) treatment, we also introduced S/T to D/A mutations to mimic the phosphorylation and de-phosphorylation WTAP in vitro and in cells, identified 5ST-D mutant as SLPS mutant, and 5ST-A mutant as LLPS mutant. We then performed 1,6-hex experiment to confirm the importance of phase separation for WTAP function, and revealed that 5ST-D SLPS mutant and 5ST-A LLPS mutant had different influence on WTAP-promoter region interaction and co-transcriptional m6A modification. Following the reviewer's suggestion, we need to further clarify the phosphorylation of WTAP phase separation. We plan to repeat the experiments by introducing potent PP4 inhibitor, fostriecin, and performed further experiments to explore the effect of WTAP IDR domain, which is reported to play a critical role for its phase separation.

1,6-hex was initially considered as the inhibitor of hydrophobic interaction which involved in various kinds of protein-protein interaction, indicating that off-target effects of 1,6-hex was inevitable. It is reported that 1,6-hex impaired RNA pol II CTD specific phosphatase and kinase activity at 5% concentration³. However, 1,6-hex is still widely used in the LLPS-associated functional studies despite its off-target effect. Related to this article, 10% 1,6-hex was reported to dissolve WTAP phase separation droplets². Beside WTAP, 1,6-hex (5%-10% w/v) was also used to explore the phase separation characteristic and function on phosphorylated protein or even kinase, including p-tau441, TAZ, HSF1 and so on⁴⁻⁶. 10% 1,6-hex inhibited the crucial role of phosphorylation-driven HSF1 LLPS in chromatin binding and transcriptional process presented by RNA-seq dataset⁶, indicating the function on kinase or phosphatase of 1,6-hex might not a global effect. To avoid the 1,6-hex-mediated kinase/phosphatase impairment in this project, we introduced the WTAP SLPS mutation and LLPS mutation besides 1,6-hex treatment to explore the m6A modification function of WTAP phase transition. We plan to repeat the experiments by lower the 1,6-hex concentration, check the WTAP phosphorylation status after 1,6-hex treatment, and discuss them in the discussion part.

A considerable number of proteins undergo phase separation via interactions between intrinsically disordered regions (IDRs). IDR contains more charged and polar amino acids to present multiple weakly interacting elements, while lacking hydrophobic amino acids to show flexible conformations⁷. In our article, we used PLAAC websites (<http://plaac.wi.mit.edu/>) to predict IDR domain of WTAP, and a fragment (234-249 amino acids) was predicted as prion-like domain. However, deletion of this fragment failed to abolish the phase separation properties of WTAP, which might be the main confusion to reviewers. To explain this issue, we checked the WTAP structure (within part of MTC complex) from protein data bank (<https://www.rcsb.org/structure/7VF2>) and found that prediction of IDR has been renewed due to the update of different algorithm. IDR of WTAP has expanded to 245-396 amino acids, containing the whole CTD region. According to our results, lack of CTD inhibited WTAP liquid-liquid phase separation both in vitro and in cells, while the phosphorylation status on CTD had dramatic impact on WTAP phase transition, which was consistent with the LLPS-regulating function of IDR. Therefore, we will revise our description on WTAP IDR, and performed further experiment to test its function.

Taken together, given the highly association between WTAP phosphorylation with phase separation status and its function during IFN- β stimulation, it is necessary to involve WTAP

phase separation in our mechanism. We will perform further experiments to propose more convincing evidence and perfect our project.

Reviewer #2 (Public review):

In this study, Cai and colleagues investigate how one component of the m6A methyltransferase complex, the WTAP protein, responds to IFN β stimulation. They find that viral infection or IFN β stimulation induces the transition of WTAP from aggregates to liquid droplets through dephosphorylation by PPP4. This process affects the m6A modification levels of ISG mRNAs and modulates their stability. In addition, the WTAP droplets interact with the transcription factor STAT1 to recruit the methyltransferase complex to ISG promoters and enhance m6A modification during transcription. The investigation dives into a previously unexplored area of how viral infection or IFN β stimulation affects m6A modification on ISGs. The observation that WTAP undergoes a phase transition is significant in our understanding of the mechanisms underlying m6A's function in immunity. However, there are still key gaps that should be addressed to fully accept the model presented.

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b. The distribution of m6A modification across the ISGs. topology

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In addition, since the authors propose a novel mechanism where the interaction between phosphorylated STAT1 and WTAP directs the MTC to the promoter regions of ISGs to facilitate co-transcriptional m6A modification, it is critical to analyze whether the m6A modification distribution holds true in the data.

We appreciate the reviewer's summary of our manuscript and the constructive assessment. We plan to perform the related analysis accordingly to present the m6A modification in ISGs in our model.

(2) Since a key part of the model includes the cytosol-localized STAT1 protein undergoing phosphorylation to translocate to the nucleus to mediate gene expression, the authors should focus on the interaction between phosphorylated STAT1 and WTAP in Figure 4, rather than the unphosphorylated STAT1. Only phosphorylated STAT1 localizes to the nucleus, so the presence of pSTAT1 in the immunoprecipitate is critical for establishing a functional link between STAT1 activation and its interaction with WTAP.

We plan to repeat the immunoprecipitation experiments to clarify the function of pSTAT1 in WTAP interaction and m6A modification as the reviewer suggested.

(3) The authors should include pSTAT1 ChIP-seq and WTAP ChIP-seq on IFN β -treated samples in Figure 5 to allow for a comprehensive and unbiased genomic analysis for comparing the overlaps of peaks from both ChIP-seq datasets. These results should further support their hypothesis that WTAP interacts with pSTAT1 to enhance m6A modifications on ISGs.

We first performed the MeRIP-seq and RNA-seq and explored the critical role of WTAP in ISGs m6A modification and expression. By immunoprecipitation and immunofluorescence

experiments, we found phase transition of WTAP enhanced its interaction to pSTAT1. These results indicate that WTAP mediated ISGs m6A modification and expression by enhanced its interaction with pSTAT1 during virus infection and IFN- β stimulation. However, we were still not sure how WTAP-mediated m6A modification related to pSTAT1-mediated transcription. By analyzing METTL3 ChIP-seq data or caPAR-CLIP-seq data, several researches have revealed the recruitment of m6A methylation complex (MTC) to transcription start sites (TSS) of coding genes and R-loop structure by interacting with transcriptional factors STAT5B or DNA helicase DDX21, indicating the engagement of MTC mediated m6A modification on nascent transcripts at the very beginning of transcription 8-10. Thus, we proposed that phase transition of WTAP could be recruited to the ISGs promoter region by pSTAT1, and verified this hypothesis by pSTAT1/WTAP-ChIP-qPCR. We believe ChIP-seq experiment is a good idea to explore the mechanism in depth, but the results in this article for now are enough to explain our mechanism. We will continuously focus on the whole genome chromatin distribution of WTAP and explore more functional effect of transcriptional factor-dependent WTAP-promoter region interaction in t.

Minor points:

- (1) Since IFN β is primarily known for modulating biological processes through gene transcription, it would be informative if the authors discussed the mechanism of how IFN β would induce the interaction between WTAP and PPP4.*
- (2) The authors should include mCherry alone controls in Figure 1D to demonstrate that mCherry does not contribute to the phase separation of WTAP. Does mCherry have or lack a PLD?*
- (3) The authors should clarify the immunoprecipitation assays in the methods. For example, the labeling in Figure 2A suggests that antibodies against WTAP and pan-p were used for two immunoprecipitations. Is that accurate?*
- (4) The authors should include overall m6A modification levels quantified of GFPsgRNA and WTAPsgRNA cells, either by mass spectrometry (preferably) or dot blot.*

We thank reviewer for raising these useful suggestions. We will perform related experiments and revised the manuscript carefully the as reviewer suggested.

Reviewer #3 (Public review):

Summary:

This study presents a valuable finding on the mechanism used by WTAP to modulate the IFN- β stimulation. It describes the phase transition of WTAP driven by IFN- β -induced dephosphorylation. The evidence supporting the claims of the authors is solid, although major analysis and controls would strengthen the impact of the findings. Additionally, more attention to the figure design and to the text would help the reader to understand the major findings.

Strength:

The key finding is the revelation that WTAP undergoes phase separation during virus infection or IFN- β treatment. The authors conducted a series of precise experiments to uncover the mechanism behind WTAP phase separation and identified the regulatory role of 5 phosphorylation sites. They also succeeded in pinpointing the phosphatase involved.

Weaknesses:

However, as the authors acknowledge, it is already widely known in the field that IFN and viral infection regulate m6A mRNAs and ISGs. Therefore, a more detailed discussion could help the reader interpret the obtained findings in light of previous research.

It is well-known that protein concentration drives phase separation events. Similarly, previous studies and some of the figures presented by the authors show an increase in WTAP expression upon IFN treatment. The authors do not discuss the contribution of WTAP expression levels to the phase separation event observed upon IFN treatment. Similarly, METTL3 and METTL14, as well as other proteins of the MTC are upregulated upon IFN treatment. How does the MTC protein concentration contribute to the observed phase separation event?

How is PP4 related to the IFN signaling cascade?

In general, it is very confusing to talk about WTAP KO as WTAPgRNA.

We are grateful for the positive comments and the unbiased advice by reviewer. To interpret the findings in previous research, we will revise the manuscript carefully and perform more detailed discussion on ISGs m6A modification during virus infection or IFN stimulation. As previously reported, WTAP protein level will be induced by long time IFN- β stimulation or LPS stimulation, while LPS-induced WTAP expression promoted its phase separation ability^{2,11}. Although there was no significant upregulation of WTAP expression level in our short time treatment, we hypothesized that WTAP phase separation will be promoted due to higher protein concentration after long time IFN stimulation, enhancing m6A modification deposition on ISGs mRNA, revealing a feedback loop between WTAP phase separation and m6A modification during specific stimulation. To discuss the effect of MTC protein concentration in our proposed event, we will perform immunoblotting experiments of MTC proteins and check the phase separation effect in different WTAP concentration.

Protein phosphatase 4 (PP4) is a multi-subunit Ser/Thr phosphatase complex that participate in diverse cellular pathways including DDR, cell cycle progression, and apoptosis¹². Protein phosphatase 4 catalytic subunit 4C (PPP4C) is one of the components of PP4 complex. Previous research showed that knockout of PPP4C enhanced IFN- β downstream signaling and gene expression, which was consistent with our findings that knockdown of PPP4C impaired WTAP-mediated m6A modification, enhanced the ISGs expression. Since there was no significant enhancement in PPP4C expression level during IFN- β stimulation in our results, we will consider to explore the post-translation modification that may influence the protein-protein interaction, such as ubiquitination.

In this project, all the WTAP-deficient THP-1 cells were bulk cells treated with WTAPsgRNA, but not monoclonal knockout cells. We confirmed that WTAP expression was efficiently knockdown in WTAPsgRNA THP-1 cells, and the m6A modification level has been impaired, avoiding the compensatory effect on m6A modification by other possible proteins. Thus, we prefer to call it WTAPsgRNA THP-1 cells rather than WTAP KO THP-1 cells.

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